SOFTWARE ENGINEERING

LAB 1

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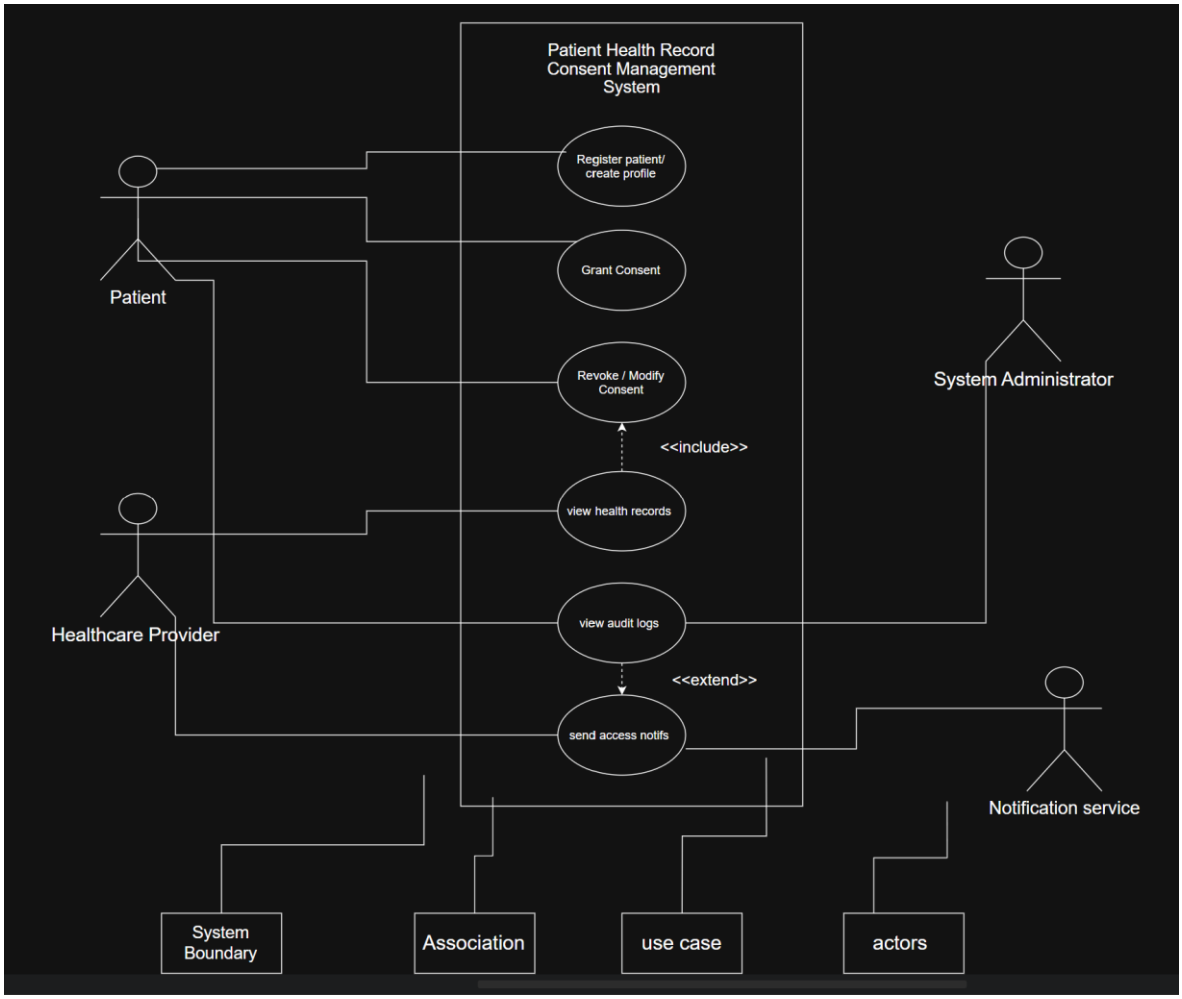
SECTION: 5B

Deliverables 1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow patients to create an account and set up their consent preferences for sharing health records.	HIGH	A patient can successfully register and view their consent dashboard after entering valid details.	Patients need an account before they can manage access to their health information.
FR-002	Functional	The system shall allow patients to grant healthcare providers permission to access specific health records.	HIGH	When a patient grants consent, the selected healthcare provider can access only the records approved by the patient.	Gives patients control over who can view their medical information.
FR-003	Functional	The system shall allow patients to withdraw or modify consent whenever they choose.	HIGH	Any changes to consent are reflected immediately, and unauthorized providers can no longer access restricted records.	Patients should be able to change their decisions as their healthcare needs change.
FR-004	Functional	The system shall keep a record of all consent-related activities, including approvals,	MEDIUM	Every consent action is stored with the user's identity, date,	Maintains transparency and supports compliance

		withdrawals, and access requests.		time, and action performed.	with healthcare regulations.
FR-005	Functional	The system shall notify patients whenever a healthcare provider accesses their health records using active consent.	MEDIUM	A notification is sent to the patient within one minute of a successful record access.	Helps patients stay informed about how their data is being used.
NFR-001	Non-Functional	The system shall process consent requests and updates quickly to provide a smooth user experience.	HIGH	At least 95% of consent actions are completed within 3 seconds.	Fast response times improve usability and user satisfaction.
NFR-001	Non-Functional	The system shall protect patient data by encrypting all health records and consent information during storage and transmission.	HIGH	Security testing confirms that encrypted communication and secure data storage are enabled.	Health records contain sensitive information that must remain confidential.

Deliverables 2. UML Use-Case Diagram



Deliverables 3. Use-Case Flow Document

Use Case: Monitor Patient Vitals and Generate Alerts

Main Success Scenario

1. The patient opens the app and signs in.
2. The wearable device automatically sends the patient's latest vital signs, such as heart rate, blood pressure, and oxygen level.
3. The system receives and saves the data securely.
4. The system checks the readings against the safe limits set by the healthcare provider.
5. All vital signs are within the normal range.
6. The patient's dashboard is updated with the latest health information.
7. The healthcare provider can view the updated readings in real time.
8. The monitoring process is completed successfully.

Alternate Flow 1: Abnormal Vital Signs Detected

4a. The system detects that one or more vital signs are outside the safe range.

- 4a1. The system immediately generates an alert. A notification is sent to the healthcare provider and caregiver.
- 4a2. The alert is saved in the system's alert history.
- 4a3. The healthcare provider reviews the patient's condition and decides on the next steps.
- 4a4. The monitoring process continues after the alert is handled.

Alternate Flow 2: Device Connection Problem

2a. The wearable device is unable to send the patient's vital signs.

- 2a1. The system detects the connection issue.
- 2a2. A message is displayed asking the patient to reconnect the device.
- 2a3. The system attempts to reconnect automatically.
- 2a4. If the connection is restored, the monitoring process continues normally.
- 2a5. If the connection cannot be restored, the patient is notified and monitoring is temporarily paused until the device is connected again.